CS 639: Intro to Foundation Models
Homework 2

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Part A: Core Transformer Implementation

(A5) The training and validation loss curves were obtained as below -

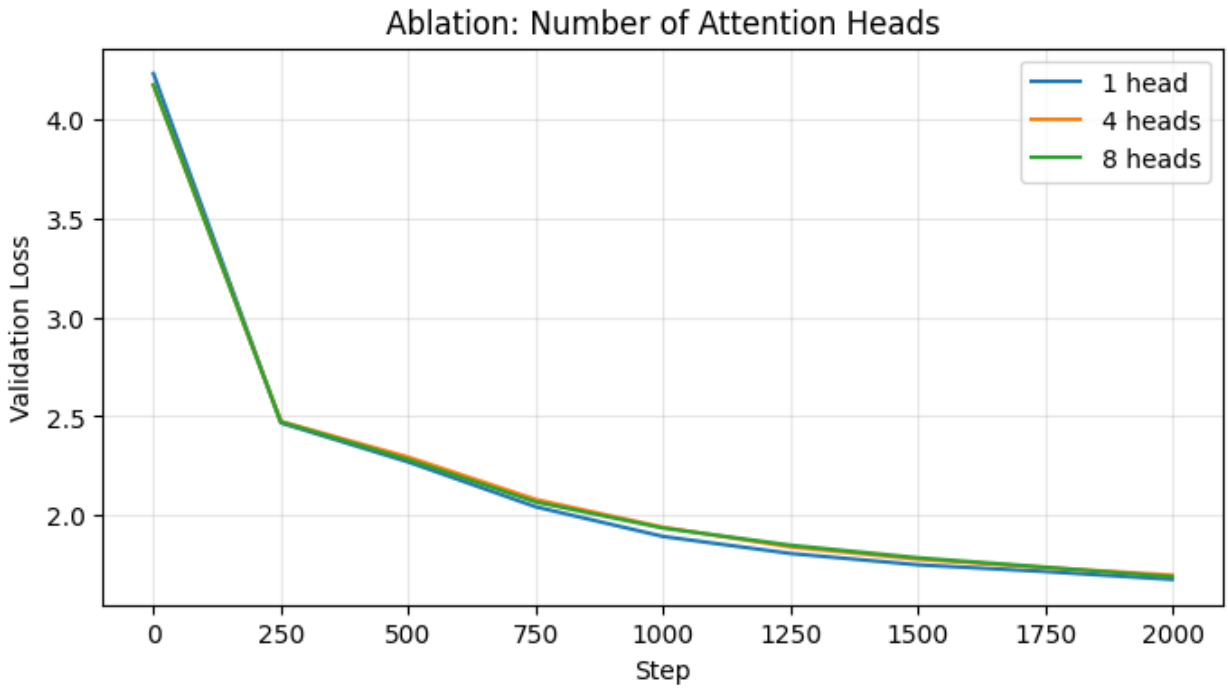
The model generated the following sample text -

```
ISABELLA:
Why, is the prince with you speak.
So this hand the lad on grans to them that
Thy thight brother hath made of my betweet,
By the peace of this their whom of Warwick,
'With these death that beding for the crown,
And the kink to se proud by to bounds well.
```

```
TYORK:
The parin's proud be stop'd good in His revenge,
And thou flatter's be foltaly lifely?

KING HENRY VI


WARWICK:
How thy desire it, and at for seem shall wash
To that when to his dear?

Sewn:
Ay, sin! I have loved my brother w
```

Part B: Ablation Runs

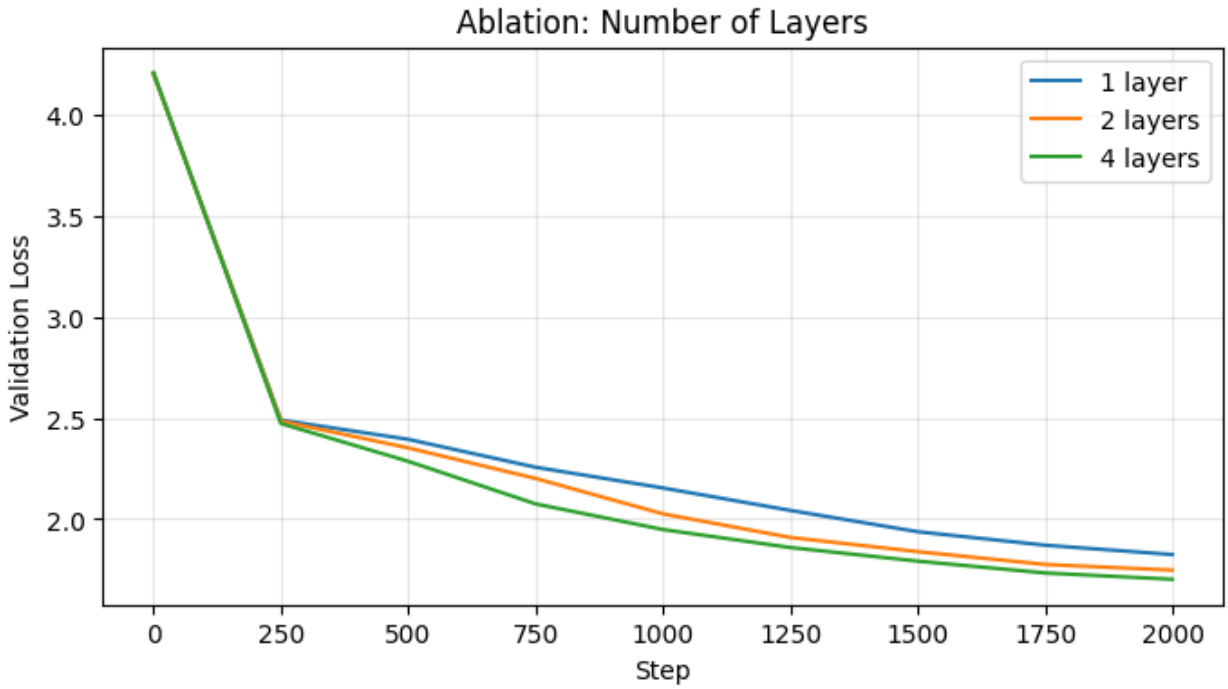
(B1) The validation loss plot was obtained as below -



The final validation loss for each configuration are as follows -

Attention heads	**Val. Loss**
1 head	1.6775
4 heads	1.6993
8 heads	1.6907

(B2) The validation loss plot was obtained as below -



The final validation loss for each configuration are as follows -

Number of Layers	**Val. Loss**
1 layer	1.8243
2 layers	1.7470
4 layers	1.7018

B1 vs B2 Comparison

We observe that the number of layers had a substantially larger effect on model performance than the number of attention heads at this scale. This could be because greater depth allows for greater abstraction. Each additional layer performs a new transformation on the data. While attention heads process information in parallel to find different types of relationships, layers process information sequentially, allowing the model to build more complex, non-linear representations of the input.

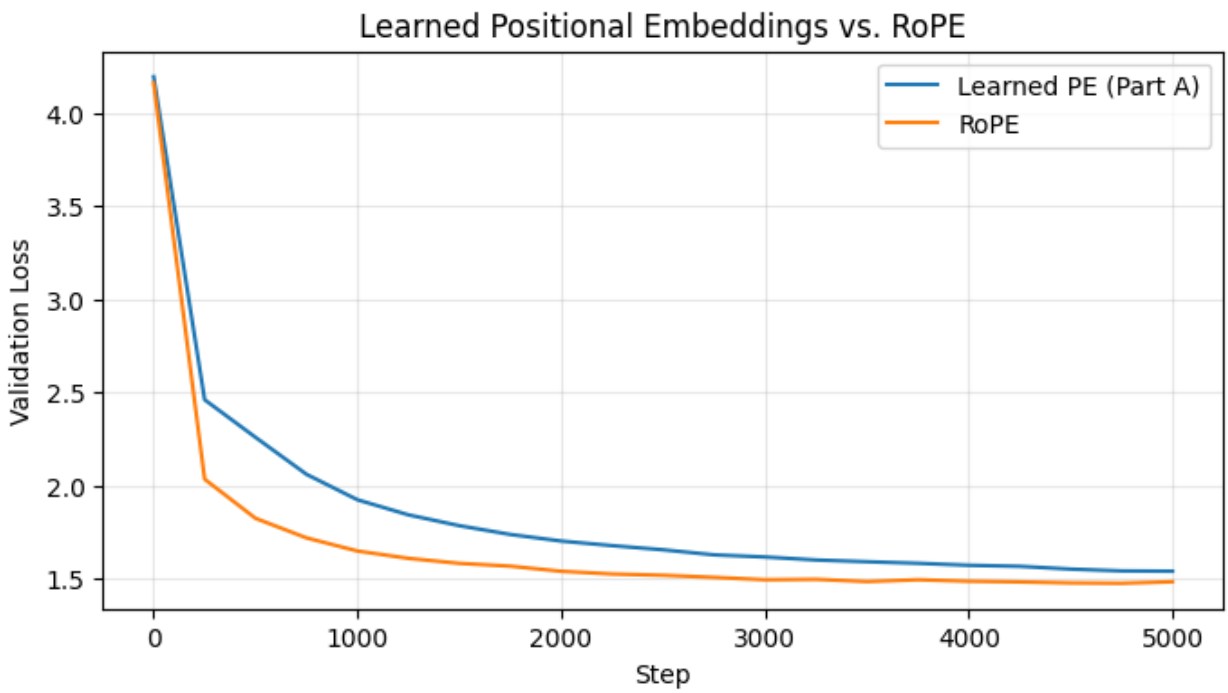
Part C: Tokenization and Positional Encoding

(C1) The two tokenization schemes compare as follows -

	Character-level	**Word-level**
Vocabulary size	65	25,670
Avg. tokens per training example	256	256
Total tokens in dataset	1,115,394	202,651
Model parameters	842,817	7,423,302
Final val loss	1.5462	8.8055

Directly comparing the raw validation loss of a character-level model to a word-level model does not make sense because cross-entropy loss is dependent on the vocabulary size and the amount of information contained in a single token. Since a character-level model only chooses from a small pool (like 65 characters), its entropy is naturally lower than a word-level model that must choose from tens of thousands of options, which mathematically forces the word-level loss to be much higher.

(C2) The combined validation loss plot comparing learned PE vs RoPE is as below -



The model generated the following sample text -

```
Than my wife staff little father: be he will build
And tnot forward to be black? then he, modth!
Alack, break the first pain shortle lease,
```

```
That's not his restored: this is a marble,
For this sin, against the poor switting.

GLOUCESTER:
Ay, madam, and hath not unfold in his sword,
Where he did content his cheeking's globle forth,
County degreen pastient.

ROMEO:
Nay, sir.

SICINIUS:
Here he heaven does me along; and in fast shows beheld
and child that doth kill him all the people.

AUTOLYCUS:
Le
```

The final validation losses are -

Positonal Encoding	**Val. Loss**
Learned PE	1.5400
RoPE	1.4830

Learned positional embeddings assign a separate trainable vector to each position in the sequence, which lets the model adapt positional representations to the training data. However, this limits the positional encodings to the maximum sequence length seen during training. RoPE instead encodes position mathematically by rotating query and key vectors using sinusoidal functions, so positional information is continuous and depends on relative distances between tokens. This allows RoPE to generalize better to longer sequences and capture relative positional relationships during attention.